FINITE LOCALLY FREE MORPHISMS ARE EXACTLY THE FINITE FLAT MORPHISMS LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A finite ring map is of finite presentation. Algebra, Lemma \ref{algebra-lemma-finite-finite-type}, cited at proof lines 12398-12399. Gives (1) implies (2).
- A finite and finitely presented algebra is finitely presented as a module. Algebra, Lemma \ref{algebra-lemma-finite-finitely-presented-extension}, cited at proof lines 12402-12404. Reduces the converse to finite projective modules.
- A finitely presented flat module is finite locally free. Algebra, Lemma \ref{algebra-lemma-finite-projective}, cited at proof lines 12404-12406. Gives (2) implies (1).
- Over a Noetherian ring a finite module is finitely presented. Algebra, Lemma \ref{algebra-lemma-Noetherian-finite-type-is-finite-presentation}, cited at proof lines 12406-12408. Gives the locally Noetherian equivalence with (3).

Source reference. Original lemma 02KB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3366–3366. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0D46: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16441–16441. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0564: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6697–6697. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00FP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 579–579. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

(1) *f is finite locally free,*
(2) *f is finite, flat, and locally of finite presentation.*

If S is locally Noetherian these are also equivalent to

(3) *f is finite and flat.*

Proof. Let $V \subset S$ be affine open. In all three cases the morphism is affine hence $f^{-1}(V)$ is affine. Thus we may write $V = \mathrm{Spec}(R)$ and $f^{-1}(V) = \mathrm{Spec}(A)$ for some R-algebra A. Assume (1). This means we can cover S by affine opens $V = \mathrm{Spec}(R)$ such that A is finite free as an R-module. Then $R \to A$ is of finite presentation by Algebra, Lemma 0D46. Thus (2) holds. Conversely, assume (2). For every affine open $V = \mathrm{Spec}(R)$ of S the ring map $R \to A$ is finite and of finite presentation and A is flat as an R-module. By Algebra, Lemma 0564 we see that A is finitely presented as an R-module. Thus Algebra, Lemma 00NX implies A is finite locally free. Thus (1) holds. The Noetherian case follows as a finite module over a Noetherian ring is a finitely presented module, see Algebra, Lemma 00FP. □